

Assignment-5

Rail Fence:-

- 1) Encrypt the message "Enjoy your journey" using Rail Fence with depth 2 & depth 3
 → Depth 2:-

Plaintext: Enjoy your journey, Ciphertext: ?

E	J	Y	O	R	O	R	E
	N	O	Y	U	J	U	N

Cipher text: EJYORORENOYUJUNY

Depth 3:-

E		Y		R		R	
	N	O	Y	U	J	U	N
		J		O		O	E

Cipher text: EYRRNOYUJUNYJOOE

- 2) Encrypt the message as per your choice using Rail Fence with depth 2 & 3
 → Plaintext: ANSHUL
 Depth 2:-

A	S	U
	N	H
		L

Ciphertext:- ASUNHL

Depth 3:-

A		U
	N	H
		L
	S	

Ciphertext:- AUNHLS

- 3) Decrypt the cipher "EOCSNYUWLFJYREASONS" encrypted using Rail Fence with depth 4:

→ E O C S
N Y U W L E
J Y R E A S
O N S

Plaintext:- "ENJOY YOUR NEW CLASSES"

"Enjoy your new classes"

- 4) Decrypt the cipher "BOOIGUYUBOSNUUTYRKAS", encrypted using Rail Fence with depth 3:

→ B O O I G
U Y U B O S N U U T
Y R K A S

Plaintext:- "BUY YOUR BOOKS IN AUGUST"

Buy your books in August

- 5) Decrypt the cipher "SEMSERTISINCMOO" encrypted using Rail fence with depth 2:

→ S E M S E R T I
S I N M O O

Plaintext: SSEIMNSCEMROTOI

Permutation:-

- 1) Encrypt the following message "We are discovered" using the key "53124".

→ Message: "We are discovered"

we are discovered

key = 5 3 1 2 4

①	②	③	④	⑤
5	3	1	2	4
w	e	a	r	e
d	i	s	c	o
v	e	r	e	d

order: ③, ④, ②, ⑤, ①

asrre eie eodwdv - Ciphertext

DES

Initial Permutation:-

58	50	42	34	26	18	10	02
60	52	44	36	28	20	12	04
62	54	46	38	30	22	14	06
64	56	48	40	32	24	16	08
57	49	41	33	25	17	09	01
59	51	43	35	27	19	11	03
61	53	45	37	29	21	13	05
63	55	47	39	31	23	15	07

- 1) Using the initial permutation table, determine the output of the initial permutation of box when the input is given in hexadecimal as:

0x0002000000000001

→ Converting them to the binary

2 - 0000

0000 0000 0000 0010 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0001

1 - 0000

4 - 0100

Only two bits are '1': input bits 15 & 64 every other are 0.

64 appears in column 1, Row 4 $\rightarrow$ that is 25th entry in the table $\rightarrow$ output bits 25 input bit 64 = 1
 15 appears in Row 8, Column 7 $\rightarrow$ that is 63rd entry in the table $\rightarrow$ output bits 63 = input bit 15 = 1

So other output bits = 0

0000 0000 0000 0000 0000 0000 0000 0000 0000
 0000 0000 0000 0000 0000 0000 0010

Output :-

0x0000 0010 0000 0002

- 2) Given the 4 bit binary plaintext (M) 1101 & the key (K) 1011 Calculate the resulting ciphertext (C) using a bitwise exclusive OR (XOR) operation. Show your bit-by-bit work. Plaintext (M = 1101 (binary)), Key = 1011 (binary))

$\rightarrow$ XOR of M = 1101 & K = 1011

M = 1 1 0 1

K = 1 0 1 1

0 1 1 0

C = 0110

- 3) The input to S-box 1 is 000000. What is the output?

→ 14 4 13 1 2 15 11 8 3 10 6 12 5 9 0 7
 0 15 7 4 14 2 13 1 10 6 12 11 9 5 3 8
 4 1 14 8 13 6 2 11 5 12 9 7 3 10 5 0
 15 12 8 2 4 9 1 7 5 11 3 14 10 0 6 13

input = 000000

Check the 0th position - 14

14 - 1110